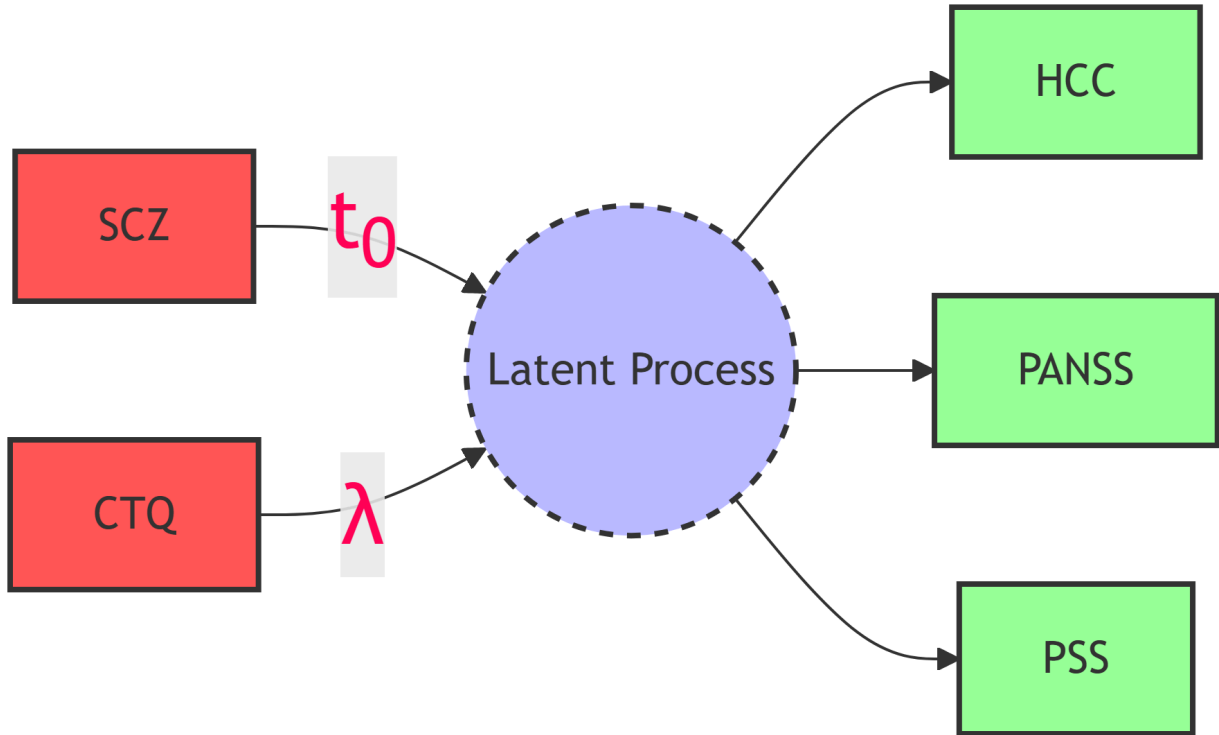


Supplementary material

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Causal structure



Model formulation

1. Latent childhood trauma metrics

For each subject i , the joint distribution of their underlying exposure to abuse ($c_{i,1}$) and neglect ($c_{i,2}$) is drawn from a centered multivariate normal distribution capturing population variance and structural trait correlation:

$$\begin{bmatrix} c_{i,1} \\ c_{i,2} \end{bmatrix} \sim \text{Multivariate-Normal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_c \right)$$

$$\Sigma_c = \text{diag}(\sigma_c) \Omega_c \text{diag}(\sigma_c)$$

Where $\sigma_c = [\sigma_{\text{abuse}}, \sigma_{\text{neglect}}]^T$ maps the population scales, and Ω_c represents the trait correlation matrix.

2. CTQ measurement (emission) layer

The individual's true latent trauma states directly manifest across the five observed subscales of the Childhood Trauma Questionnaire. The Abuse factor ($c_{i,1}$) drives emotional, physical, and sexual dimensions, while the Neglect factor ($c_{i,2}$) drives emotional and physical neglect indicators:

$$\begin{aligned} y_m^{(\text{EA})} &\sim \text{Normal} \left(\alpha_{\text{EA}} + c_{\text{subj } [m],1}, \sigma_{\text{EA}} \right) & [\text{Emotional Abuse}] \\ y_m^{(\text{PA})} &\sim \text{Normal} \left(\alpha_{\text{PA}} + c_{\text{subj } [m],1}, \sigma_{\text{PA}} \right) & [\text{Physical Abuse}] \\ y_m^{(\text{SA})} &\sim \text{Normal} \left(\alpha_{\text{SA}} + c_{\text{subj } [m],1}, \sigma_{\text{SA}} \right) & [\text{Sexual Abuse}] \\ y_m^{(\text{EN})} &\sim \text{Normal} \left(\alpha_{\text{EN}} + c_{\text{subj } [m],2}, \sigma_{\text{EN}} \right) & [\text{Emotional Neglect}] \\ y_m^{(\text{PN})} &\sim \text{Normal} \left(\alpha_{\text{PN}} + c_{\text{subj } [m],2}, \sigma_{\text{PN}} \right) & [\text{Physical Neglect}] \end{aligned}$$

3. Latent dynamics

The latent dysregulation state $z_i(t)$ transitions over irregular time intervals ($\Delta t = t - t'$) according to a continuous-time (Ornstein–Uhlenbeck) engine. The individual recovery speed λ_i forces a strict positive bound via the structural baseline floor λ_0 :

$$\begin{aligned} z_i(t) \mid z_i(t') &\sim \text{Normal} \left(e^{-\lambda_i \Delta t} z_i(t'), \sigma_z \sqrt{1 - e^{-2\lambda_i \Delta t}} \right) \\ \lambda_i &= \lambda_0 + \exp(\gamma_{\text{abuse}} c_{i,1} + \gamma_{\text{neglect}} c_{i,2}) \end{aligned}$$

4. Longitudinal process state

The state trajectory begins at a unique initial tracking baseline anchored to a matched-pair intercept (u_{pair}) and shifted by diagnostic status (SCZ):

$$\begin{aligned} z_i(t_0) &\sim \text{Normal} \left(u_{\text{pair } [i]} + \beta_{\text{SCZ}} \text{SCZ}_i, 1 \right) \\ u_j &\sim \text{Normal} \left(0, \sigma_{\text{pair}} \right) \end{aligned}$$

The evolving state $z_i(t)$ projects concurrent signals down into both historical biological archives and acute clinical cross-sections:

$$\begin{aligned} y_m^{(\text{HCC})} &\sim \text{Student-t} \left(\nu = 6, a_{\text{HCC}} + b_{\text{HCC}} z_{\text{state } [m]}, \sigma_{\text{HCC}} \right) \\ y_m^{(\text{PANSS})} &\sim \text{Normal} \left(a_{\text{PANSS}} + b_{\text{PANSS}} z_{\text{state } [m]}, \sigma_{\text{PANSS}} \right) \\ y_m^{(\text{PSS})} &\sim \text{Normal} \left(a_{\text{PSS}} + b_{\text{PSS}} z_{\text{state } [m]}, \sigma_{\text{PSS}} \right) \end{aligned}$$

5. Prior distributions

$$\begin{aligned} \lambda_0, \gamma_{\text{abuse}}, \gamma_{\text{neglect}}, \beta_{\text{SCZ}} &\sim \text{Normal} (0, 1) \\ \alpha_{\text{EA}}, \alpha_{\text{PA}}, \alpha_{\text{SA}}, \alpha_{\text{EN}}, \alpha_{\text{PN}} &\sim \text{Normal} (0, 1) \\ a_{\text{PANSS}}, a_{\text{PSS}}, a_{\text{HCC}} &\sim \text{Normal} (0, 0.5) \\ b_{\text{PANSS}}, b_{\text{PSS}}, b_{\text{HCC}} &\sim \text{Normal} (0, 1) \\ \sigma_z, \sigma_{\text{pair}}, \sigma_{\text{HCC}}, \sigma_{\text{PANSS}}, \sigma_{\text{PSS}} &\sim \text{Exponential} (1) \\ \sigma_{\text{EA}}, \sigma_{\text{PA}}, \sigma_{\text{SA}}, \sigma_{\text{EN}}, \sigma_{\text{PN}} &\sim \text{Exponential} (1) \\ \sigma_c &\sim \text{Exponential} (1) \\ \Omega_c &\sim \text{LKJ} (\eta = 2) \end{aligned}$$

Stan code

The mathematical model is rewritten in the Stan language.

```
data{
  int <lower=1> N_state;
```

```

array[N_state] int <lower=1> subj_id;
int <lower=1> N_pair;
array[N_state] int <lower=1> pair_id;
vector[N_state] time;
array[N_state] int <lower=0, upper=1> is_first;
array[N_state] int <lower=0, upper=1> scz;

int <lower=1> N_hcc;
array[N_hcc] int <lower=1> state_hcc;
vector[N_hcc] y_hcc;

int <lower=1> N_panss;
array[N_panss] int <lower=1> state_panss;
vector[N_panss] y_panss;

int <lower=1> N_pss;
array[N_pss] int <lower = 1> state_pss;
vector[N_pss] y_pss;

int <lower=1> N_ctq;
int <lower=1> N_subj;
array[N_ctq] int <lower=1> subj_ctq;

vector[N_ctq] y_ea;
vector[N_ctq] y_pa;
vector[N_ctq] y_sa;
vector[N_ctq] y_en;
vector[N_ctq] y_pn;

}

parameters{

vector[N_state] z_raw;
real <lower=0> lambda_0;
real gamma_abuse;
real gamma_neglect;
real <lower=0> sigma_z;
real <lower=0> sigma_pair;

vector[N_pair] u_pair;

real beta_scz;

real a_hcc;
real <lower=0> b_hcc;
real <lower=0> sigma_hcc;

real a_panss;
real b_panss;
real <lower=0> sigma_panss;

real a_pss;
real b_pss;
real <lower=0> sigma_pss;

```

```

real <lower=0> sigma_ea;
real <lower=0> sigma_pa;
real <lower=0> sigma_sa;
real <lower=0> sigma_en;
real <lower=0> sigma_pn;

real alpha_ea;
real alpha_pa;
real alpha_sa;
real alpha_en;
real alpha_pn;

matrix[2, N_subj] c_raw;
vector <lower=0> [2] sigma_c;
cholesky_factor_corr[2] L_c;

}

transformed parameters {

// Covariance matrix for CTQ latent factor c1 (abuse) and c2 (neglect)
matrix[2, N_subj] c;
c = diag_pre_multiply(sigma_c, L_c) * c_raw;

// latent subject-level factors c[1, ] and c[2, ]. Implicit Bayesian imputation.
vector[N_subj] lambda_subj;

for (i in 1:N_subj) {
  lambda_subj[i] = lambda_0 + exp(gamma_abuse * c[1, i] + gamma_neglect * c[2, i]);
}

// dynamic latent process
vector[N_state] z;
for (k in 1:N_state) {
  // first point in time aligned between patients and controls
  if (is_first[k] == 1) {
    z[k] = u_pair[pair_id[k]] + beta_scz * scz[k] + 1 * z_raw[k];
  // Ornstein-Uhlenbeck process (exponential return with stochastic volatility)
  } else {
    real dt = time[k] - time[k-1];
    real phi = exp(-lambda_subj[subj_id[k]] * dt);
    real sigma_dt = sigma_z * sqrt(1 - square(phi));
    z[k] = phi * z[k-1] + sigma_dt * z_raw[k];
  }
}

}

model{

// priors
lambda_0 ~ normal(0, 1);
gamma_abuse ~ normal(0, 1);

```

```

gamma_neglect ~ normal(0, 1);
sigma_z ~ exponential(1);
sigma_pair ~ exponential(1);
sigma_hcc ~ exponential(1);
beta_scz ~ normal(0, 1);
a_hcc ~ normal(0, 0.5);
b_hcc ~ normal(0, 1);
a_panss ~ normal(0, 0.5);
b_panss ~ normal(0, 1);
a_pss ~ normal(0, 0.5);
b_pss ~ normal(0, 1);
z_raw ~ std_normal();

u_pair ~ normal(0, sigma_pair);

sigma_ea ~ exponential(1);
sigma_pa ~ exponential(1);
sigma_sa ~ exponential(1);
sigma_en ~ exponential(1);
sigma_pn ~ exponential(1);

alpha_ea ~ normal(0, 1);
alpha_pa ~ normal(0, 1);
alpha_sa ~ normal(0, 1);
alpha_en ~ normal(0, 1);
alpha_pn ~ normal(0, 1);

to_vector(c_raw) ~ std_normal();

L_c ~ lkj_corr_cholesky(2);

// hcc emissions
for (m in 1:N_hcc) {
  y_hcc[m] ~ student_t(6, a_hcc + b_hcc * z[state_hcc[m]], sigma_hcc);
}

// panss emissions
for (m in 1:N_panss) {
  y_panss[m] ~ normal(a_panss + b_panss * z[state_panss[m]], sigma_panss);
}

// pss emissions
for (m in 1:N_pss) {
  y_pss[m] ~ normal(a_pss + b_pss * z[state_pss[m]], sigma_pss);
}

// CTQ emissions
y_ea ~ normal(alpha_ea + c[1, subj_ctq]', sigma_ea);
y_sa ~ normal(alpha_sa + c[1, subj_ctq]', sigma_sa);
y_pa ~ normal(alpha_pa + c[1, subj_ctq]', sigma_pa);
y_en ~ normal(alpha_en + c[2, subj_ctq]', sigma_en);
y_pn ~ normal(alpha_pn + c[2, subj_ctq]', sigma_pn);

}

```

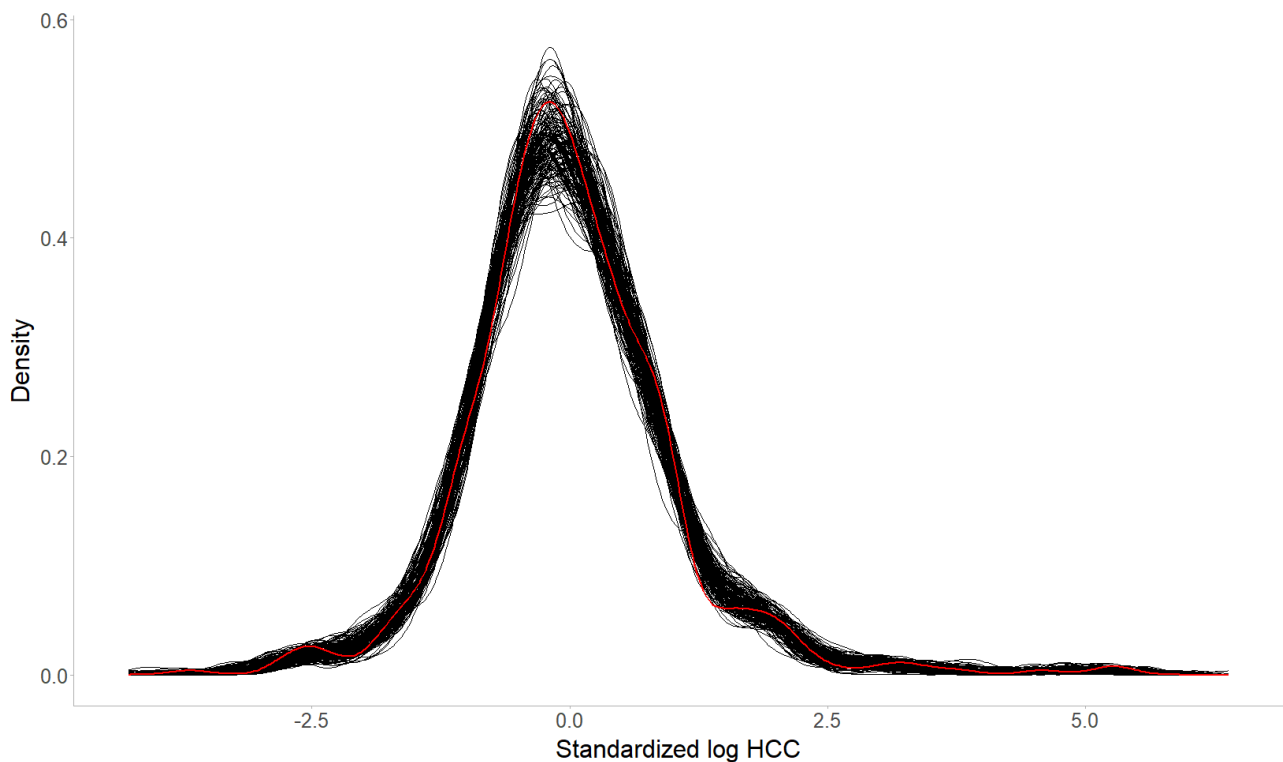
```
generated quantities{
```

```
  matrix[2, 2] corr = multiply_lower_tri_self_transpose(L_c); // Correlation matrix  
  matrix[2, 2] cov = quad_form_diag(corr, sigma_c); // (S * R * S) Covariance matrix  
}
```

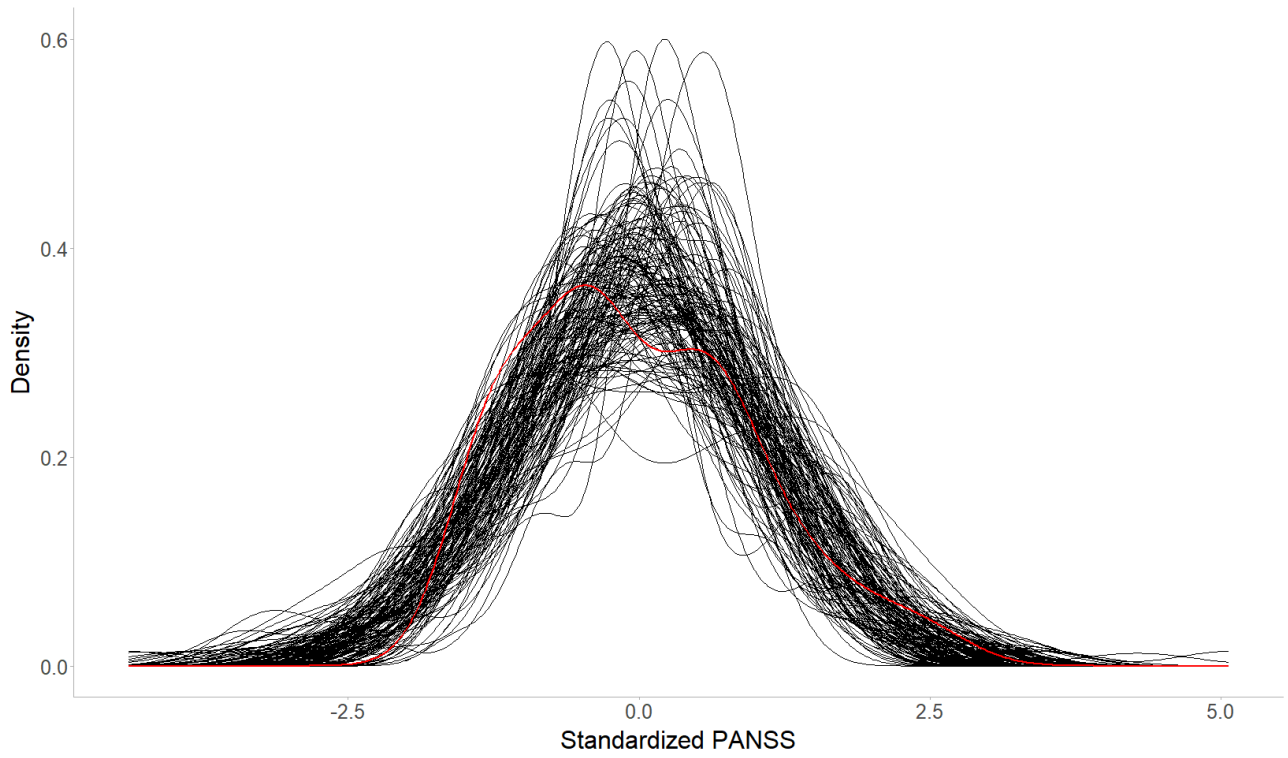
Model diagnostics and posterior predictive checks

The model was rigorously evaluated for convergence using four Markov chains with 4000 iterations each. The Rhat diagnostic was used to assess convergence, with a value below 1.01 suggesting adequate convergence. The posterior predictive checks follow.

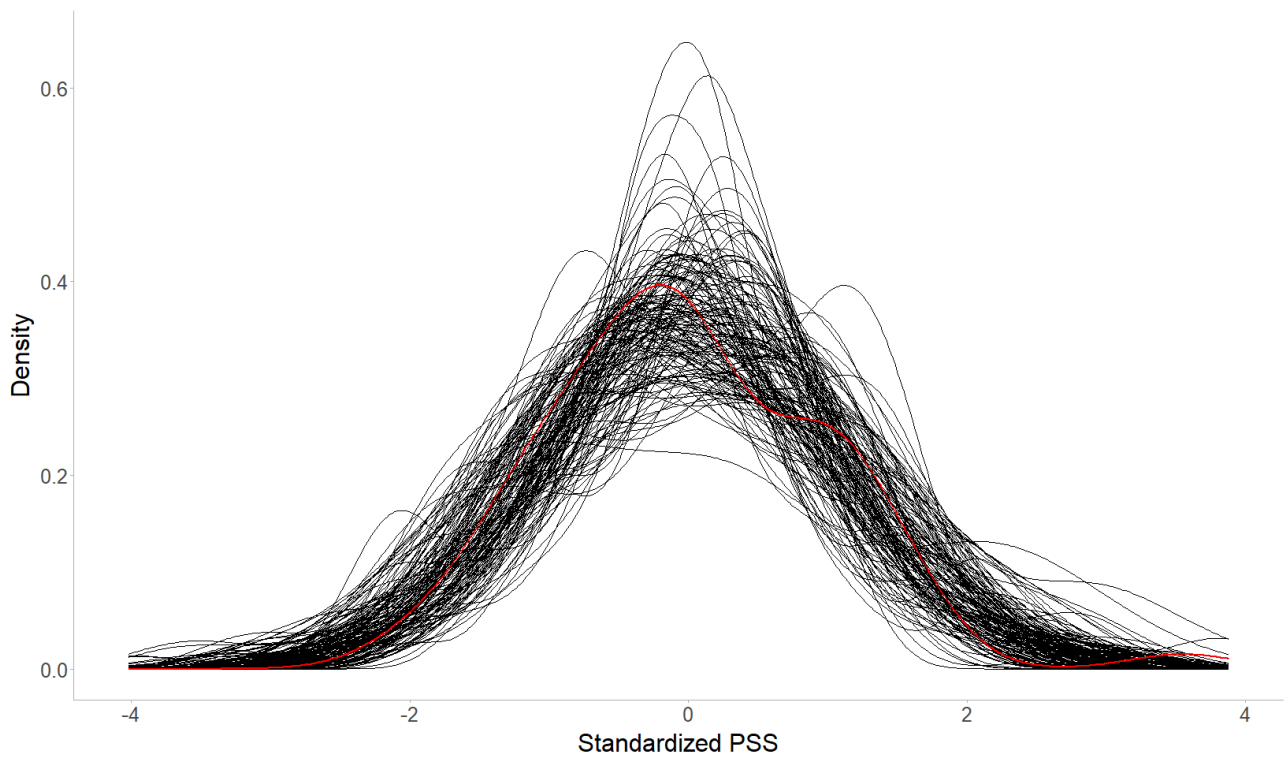
Density



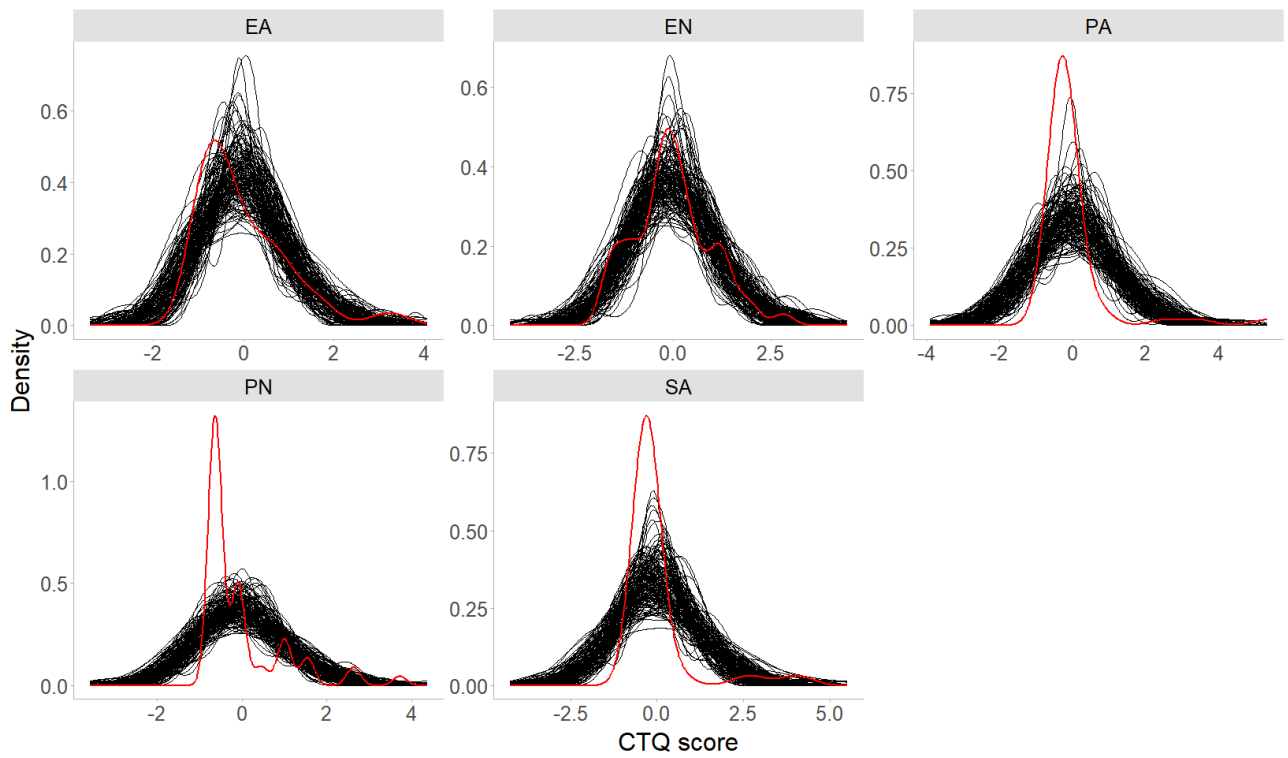
S_Figure 1: Posterior predictive check of HCC. The red line shows the observed values.



S_Figure 2: Posterior predictive check of PANSS. The red line shows the observed values.

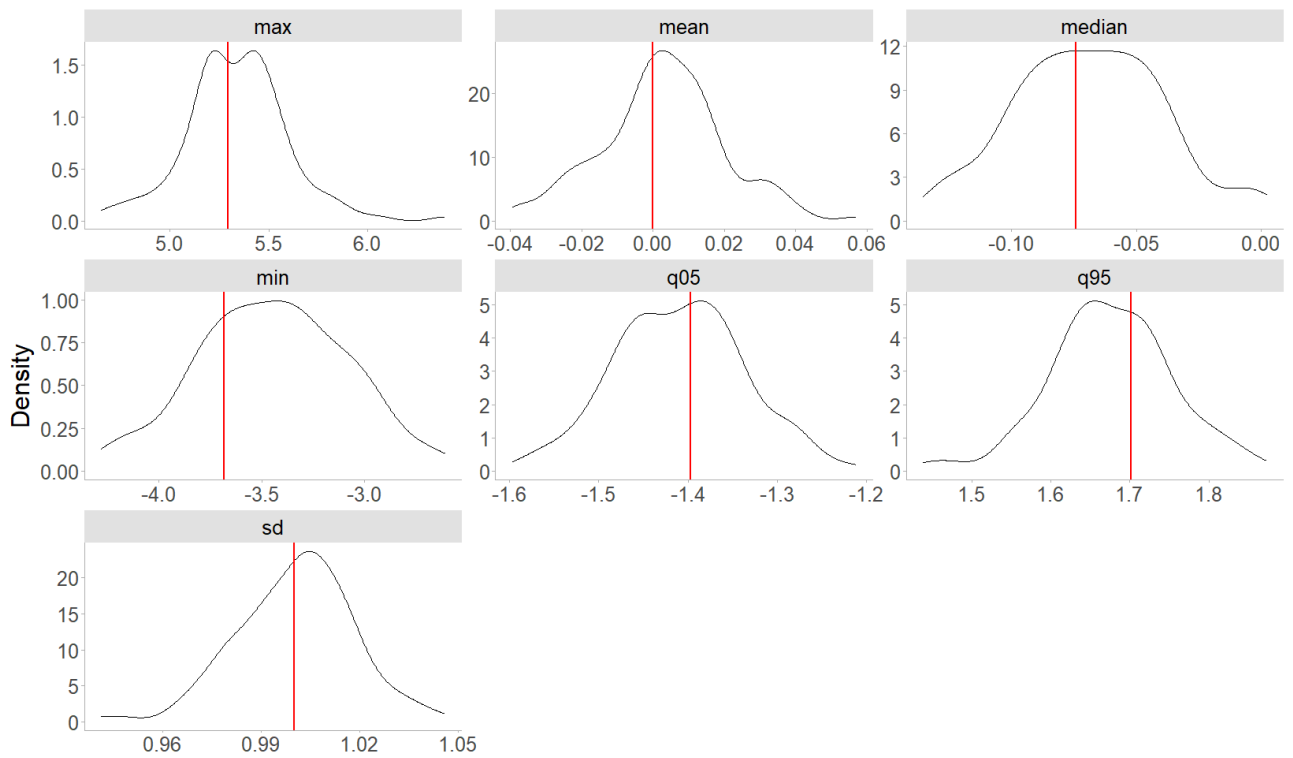


S_Figure 3: Posterior predictive check of PSS. The red line shows the observed values.

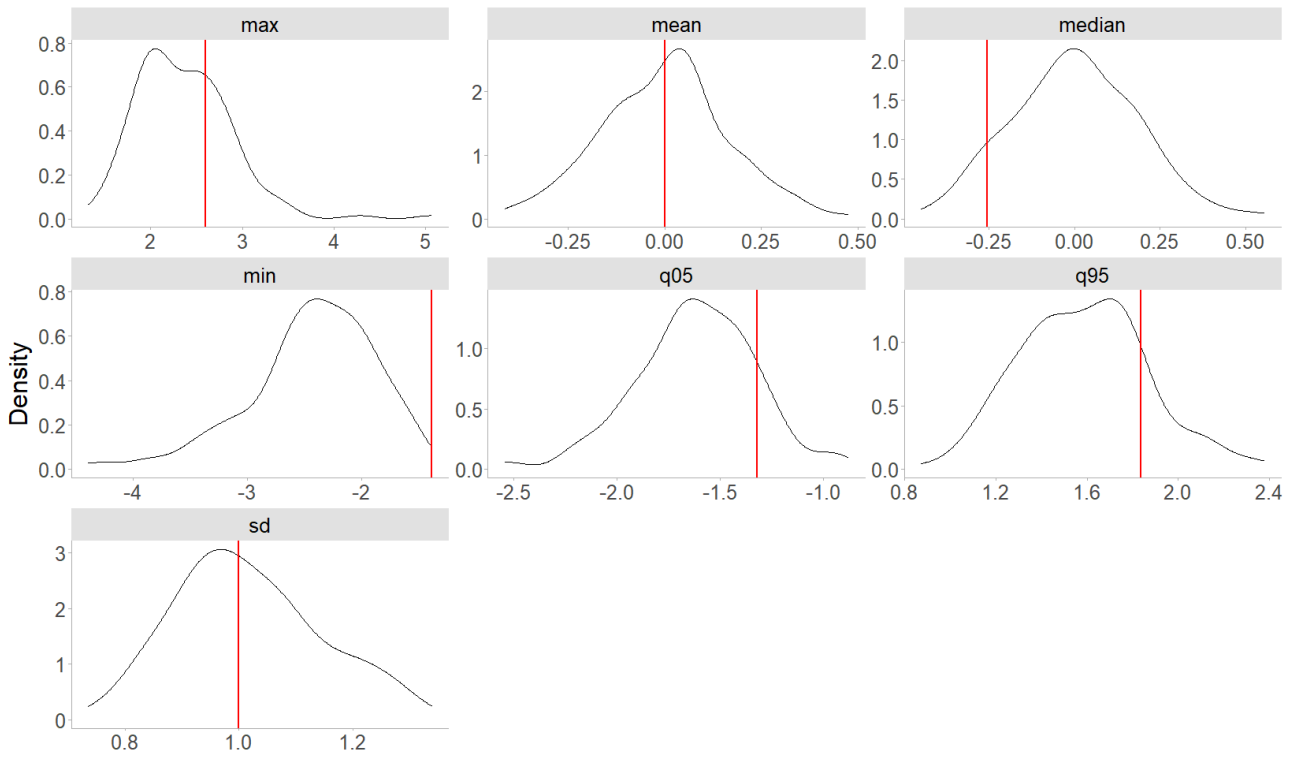


S_Figure 4: Posterior predictive check of the latent CTQ. The red line shows the observed values.

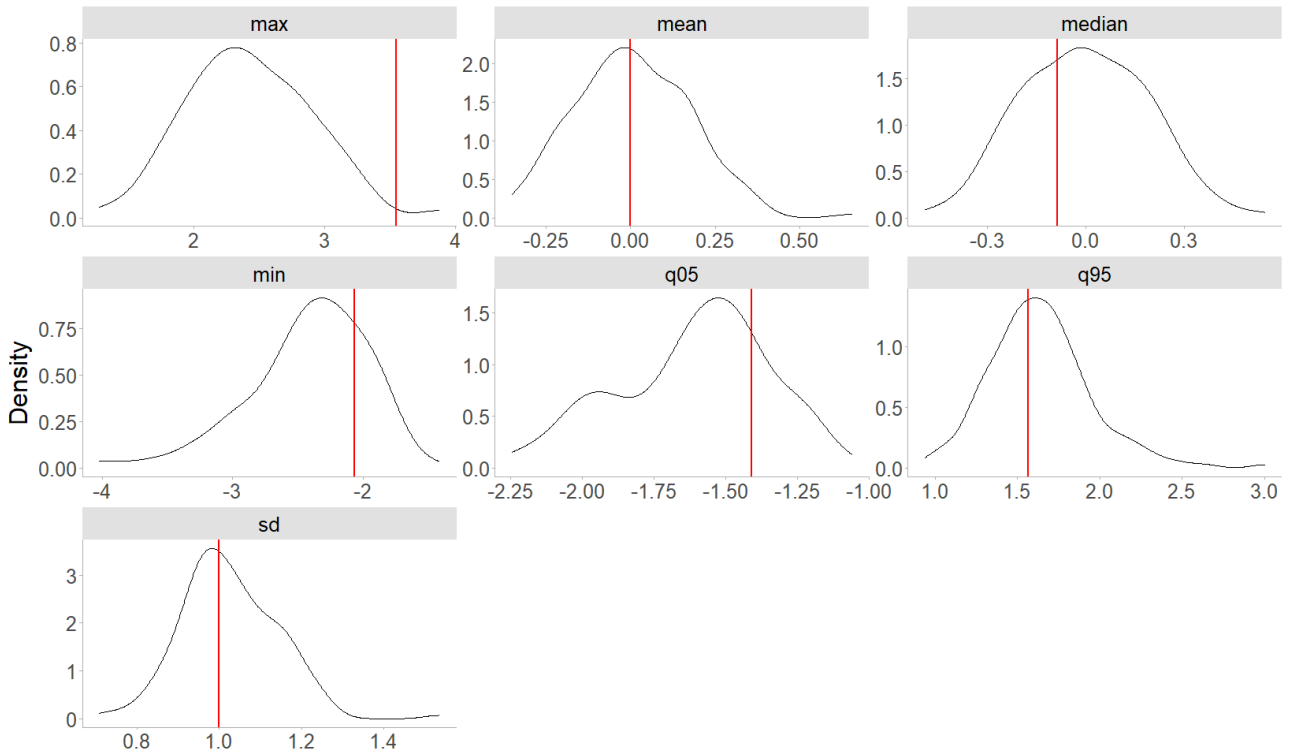
Statistics



S_Figure 5: Posterior predictive check of selected statistics of HCC. The red line shows the observed values.



S_Figure 6: Posterior predictive check of selected statistics of PANSS. The red line shows the observed values.



S_Figure 7: Posterior predictive check of selected statistics of PSS. The red line shows the observed values.

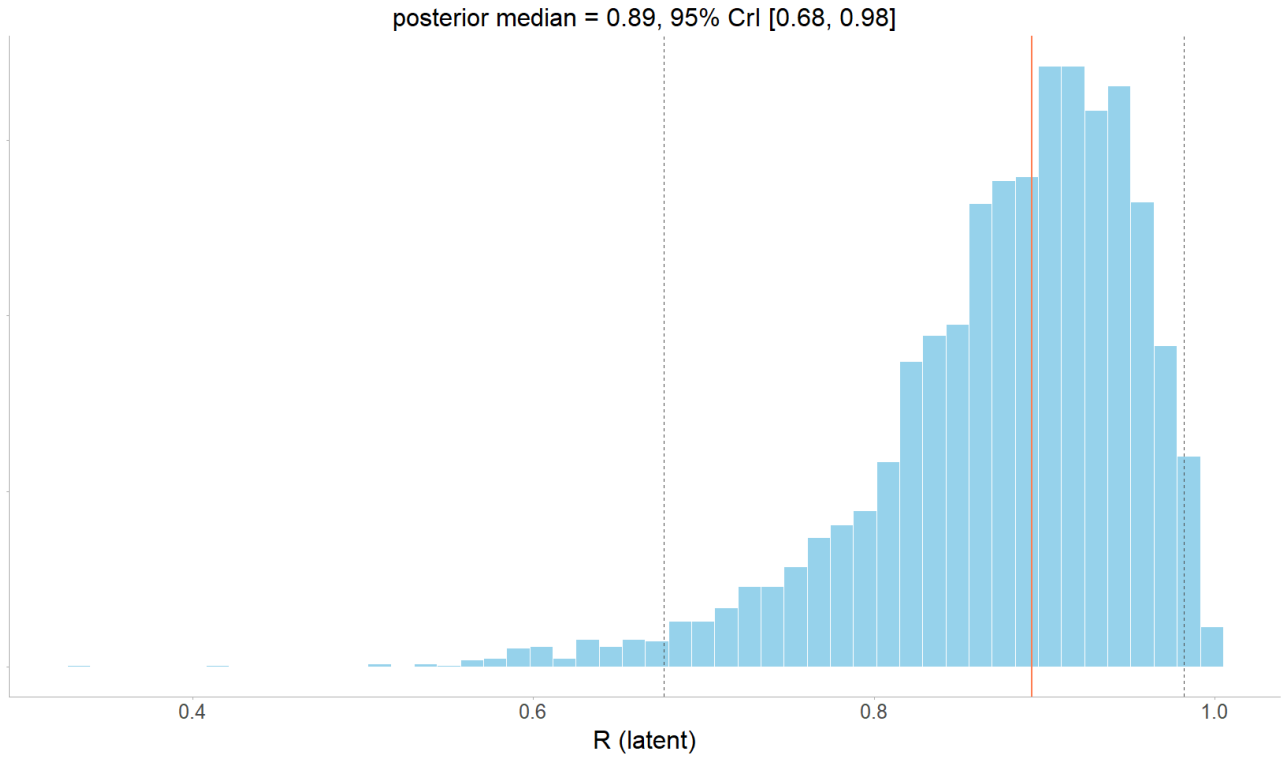
Latent trauma metric stratification

CTQ subscales exhibit high mutual inter-correlation due to shared response variance. Modeling observed subscale scores as independent predictors introduces multicollinearity and leads to overestimating trauma burden when multiple subscales are elevated simultaneously (analogous to multiple motion sensors triggering from a single event). To resolve this, we employed a Bayesian implicit latent variable framework, modeling five observed CTQ subscale indicators (y_{EA} , y_{PA} , y_{SA} , y_{EN} , y_{PN}) as expressions of two correlated latent factors: Abuse (c_1) and Neglect (c_2). Mean latent factor score (c_1 , c_2) represents standardized, continuous latent trait category, by which the subjects were assigned into trauma severity tiers.

Low trauma: $\frac{1}{2}(c_1 + c_2) \leq -0.75$

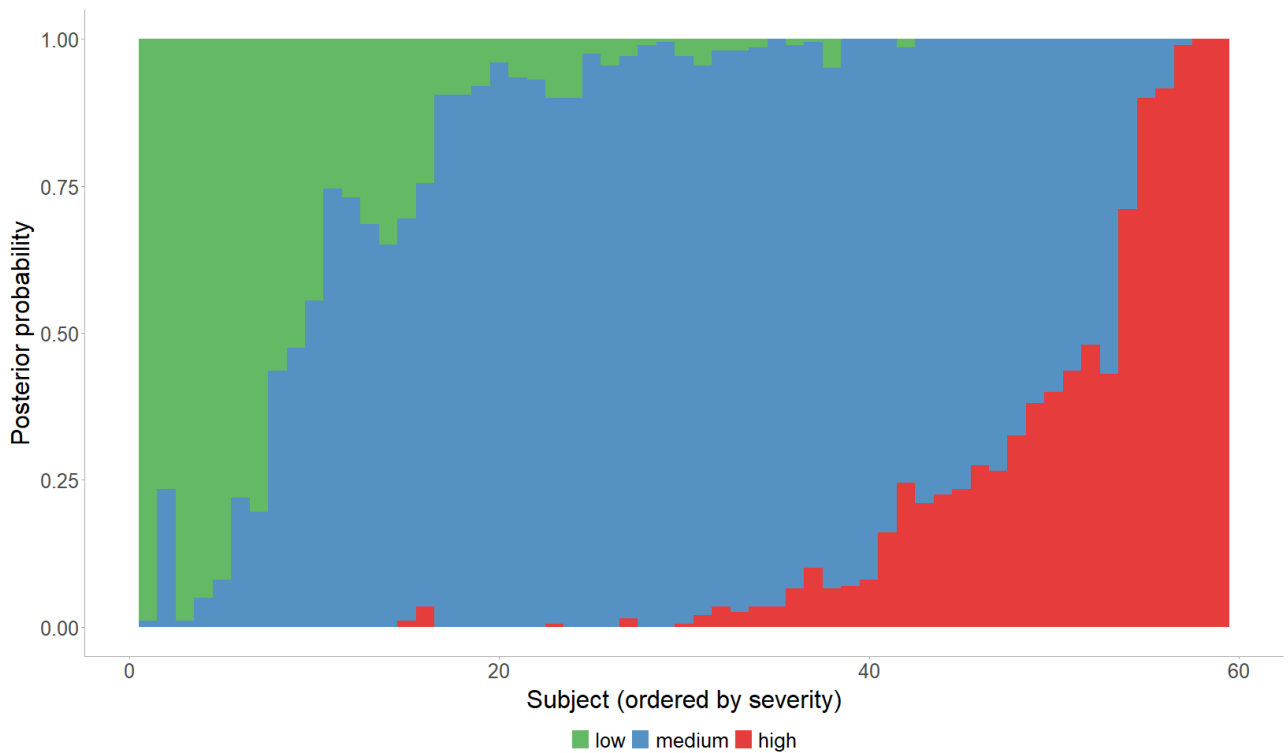
Medium trauma: $-0.75 < \frac{1}{2}(c_1 + c_2) < 0.75$

High trauma: $\frac{1}{2}(c_1 + c_2) \geq 0.75$



S_Figure 8: Latent abuse–neglect correlation. The red line shows the posterior median and the dashed lines display the 95% credibility interval.

Because $c_{1,i}$ and $c_{2,i}$ are full posterior draws (not point estimates), each subject gets a posterior distribution over tiers rather than a single label.



S_Figure 9: Posterior trauma-tier posterior classification per subject - stacked posterior probability of low / medium / high, subjects ordered by mean latent severity.

Posterior means of the latent metrics related to the CTQ native scale could be displayed as well.

